Open Economy Macroeconomics I

Macroeconomics B
Lecture 9 (Chapter 23)

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Where We Are and What Came Before

You have built the short-run model in a closed economy:

- AD: output demand falls when the relevant real interest rate rises.
- AS: inflation depends on expected inflation, the output gap, and supply shocks.
- Stabilisation policy: demand shocks can be stabilised more easily than supply shocks.
- Rational expectations: predictable policy works differently when people understand the rule.

Today we open the economy. The new state variable is **competitiveness**, measured by the real exchange rate. The new policy constraint is the **trilemma**.

Roadmap for Today

1. Financial openness and uncovered interest parity (UIP).
2. The macroeconomic trilemma.
3. Real exchange rates, competitiveness, and purchasing power parity.
4. Net exports and the Marshall-Lerner condition.
5. Mid-lecture Socrative check (three questions).
6. The open-economy IS curve and the preliminary AD-AS model.
7. Denmark and the Novo Nordisk decade as a real-world application.

Main reading: Sørensen & Whitta-Jacobsen, Chapter 23.

Learning Goals

After this lecture, **you** will be able to:

- derive and interpret uncovered interest parity in exact and log-linear form,
- explain the trilemma and identify which corner Denmark and Sweden occupy,
- define the real exchange rate and compute how inflation differentials affect competitiveness,
- state the Marshall-Lerner condition and explain the J-curve intuition,
- write the open-economy IS curve and identify the roles of β_1 and β_2,
- trace how a small open economy can absorb a demand shock through inflation differentials.

Hook: The Quietest Krone Market Since 1982

Three Danish facts that look unrelated, but are not:

- From late 2023 until the end of February 2025, Danmarks Nationalbank did not intervene in the FX market a single time. This is the longest intervention-free period since the fixed exchange rate policy began in 1982.
- Since June 2024 the Nationalbank cut its policy rate eight times in a row, each cut matching the ECB. The current-account rate stood at 1.60% in May 2025.
- Denmark's current-account surplus was about 8.5% of GDP in 2024.

Class question: What do all three facts have in common? Today's model gives you the answer.

Source: Danmarks Nationalbank, Annual Report 2024; FXEmpire, May 2025; Statistics Denmark, 2024.

What Changes When We Open the Economy?

The closed-economy model had one main demand channel: the real interest rate. A small open economy adds two more.

Channel	Key variable	Economic mechanism
Financial market	$i - i^f$	Capital flows link domestic and foreign returns.
Competitiveness	e^r	Relative prices shift demand between domestic and foreign goods.
Foreign demand	y^f	Trading partners' booms and recessions move exports.

The lecture separates the **financial channel** first, then the **trade channel**, and finally combines both into AD-AS.

The Small Open Economy: Core Assumptions

Chapter 23 starts with a deliberately simple environment.

- **Small**: domestic choices do not move foreign interest rates, foreign prices, or foreign output.
- **Open**: goods, services, and financial capital cross borders.
- **Specialised**: domestic and foreign goods are imperfect substitutes.
- **Perfect capital mobility**: investors can move funds across countries instantly.
- **Risk-neutral investors**: the baseline abstracts from risk premia.

These strong assumptions make the arbitrage condition transparent. Risk premia and frictions return later to explain deviations in the data.

Exchange Rate Notation: One Convention to Remember

Let E_t be the nominal exchange rate:

$$E_t = \text{domestic currency price of one unit of foreign currency.}$$

- For Denmark against the euro, E_t is DKK per EUR. The central rate is $E = 7.46038$.
- $E_t \uparrow$: one euro costs more kroner, so the krone **depreciates**.
- Lowercase $e_t \equiv \ln E_t$.
- Expected nominal depreciation is $\Delta e^e_{t+1} \equiv e^e_{t+1} - e_t$.

Higher e means a weaker domestic currency. This single rule fixes the sign of everything that follows.

UIP: The Exact No-Arbitrage Condition

An investor can put one unit of domestic currency into either a domestic or a foreign bond.

Domestic bond	Foreign bond
End-of-period payoff: $1 + i_t$	Convert, invest abroad, convert back: $(1 + i_t^f)\frac{E_{t+1}^e}{E_t}$

No arbitrage with risk-neutral investors requires equal expected returns:

$$\boxed{1 + i_t = (1 + i_t^f)\frac{E_{t+1}^e}{E_t}}$$

Derivation

UIP: The Log-Linear Version

Taking logs and using $\ln(1+x) \approx x$ gives the working form:

$$\boxed{i_t \approx i_t^f + \Delta e_{t+1}^e}$$

- If investors expect the domestic currency to depreciate, $\Delta e_{t+1}^e > 0$, the domestic interest rate must be higher to compensate.
- If expected depreciation is zero, $i_t \approx i_t^f$.
- This is the single equation behind most of the trilemma logic that follows.

Mechanism in one line: interest-rate differentials and expected exchange-rate movements are two sides of the same arbitrage condition.

UIP Mechanism: How Markets Restore the Parity

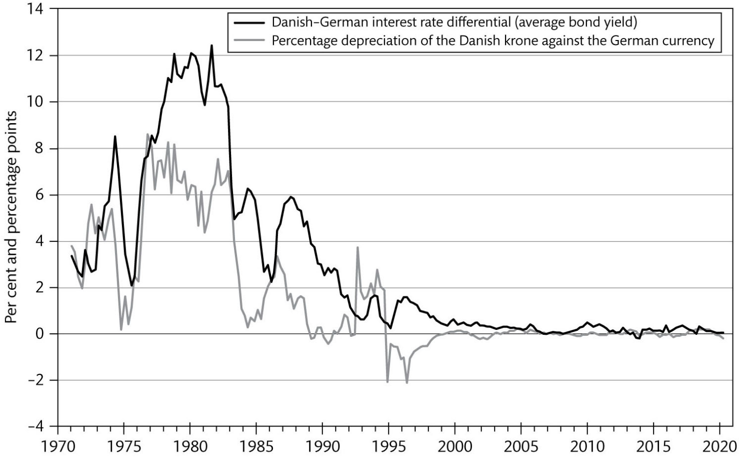
Suppose, for a moment,

$$i_t > i_t^f + \Delta e_{t+1}^e.$$

1. Domestic assets offer a higher expected return than foreign assets.
2. Investors buy domestic bonds, causing capital inflows.
3. Demand for domestic currency rises, so the spot rate E_t falls (domestic currency appreciates today).
4. With E_t lower and E_{t+1}^e unchanged, expected depreciation Δe_{t+1}^e rises until the parity is restored.

In equilibrium, no free expected return remains. Higher domestic rates are exactly offset by expected currency depreciation.

UIP in the Data: Denmark and Germany



Source: Sørensen & Whitta-Jacobsen, Figure 23.2.
Interest differentials were large when the krone peg was less credible (1970s and 1980s). They collapsed to near zero once the peg became highly credible from the 1990s onward, exactly as UIP predicts.

The Macroeconomic Trilemma

With free capital mobility, UIP makes it impossible to choose all three objectives at once. You pick two.

Chosen objectives	What follows from UIP	Example
Free capital and fixed exchange rate	$\Delta e^e = 0$, so $i \approx i^f$	Denmark, Hong Kong
Free capital and independent monetary policy	exchange rate must float	Sweden, UK, Norway
Fixed exchange rate and independent monetary policy	capital mobility must be limited	Bretton Woods, China (historically)

Denmark chose the first row. That choice ties monetary policy to the ECB and pushes adjustment onto fiscal policy, wages and prices.

Real Exchange Rate: Definition

The real exchange rate compares foreign and domestic price levels measured in the same currency:

$$\boxed{E_t^r = \frac{E_t P_t^f}{P_t}}$$

- $E_t P_t^f$: foreign price level expressed in domestic currency.
- P_t: domestic price level.
- $E_t^r \uparrow$: foreign goods become more expensive relative to domestic goods.
- Domestic competitiveness therefore **improves** when E_t^r rises.

The terms of trade move in the opposite direction: $1/E_t^r$.

Real Exchange Rate Dynamics

Take logs: $e_t^r = e_t + p_t^f - p_t$. Differencing gives the key result:

$$\boxed{\Delta e_t^r = \Delta e_t + \pi_t^f - \pi_t}$$

- Nominal depreciation, $\Delta e_t > 0$, improves competitiveness.
- Domestic inflation above foreign inflation, $\pi_t > \pi_t^f$, erodes competitiveness.
- Under a fixed exchange rate, $\Delta e_t = 0$, so the **inflation differential is the only adjustment channel**.

Derivation

Relative Purchasing Power Parity (RPPP)

Competitiveness cannot drift forever. In the long run, $\Delta e_t^r = 0$, which gives

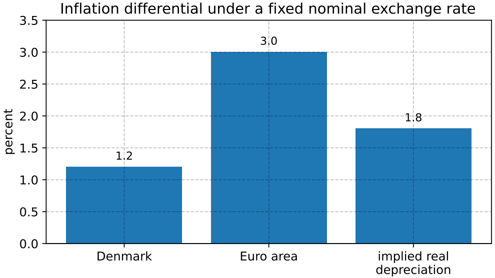
$$\boxed{\Delta e_t = \pi_t - \pi_t^f}$$

- This is relative purchasing power parity.
- It is a long-run tendency, not a period-by-period law.
- If domestic inflation is persistently higher, the nominal exchange rate must depreciate in the long run to keep competitiveness stable.
- For a fixed exchange rate, RPPP requires $\pi_t = \pi_t^f$ on average.

Denmark and the Euro Area: A Recent Inflation Differential

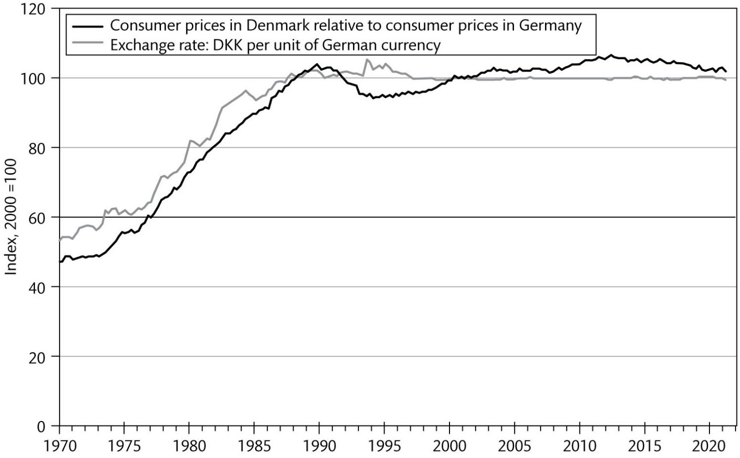
In April 2026, Danish HICP inflation was 1.2% year-on-year; euro area inflation was about 3.0%.
With $\Delta e_t \approx 0$ under the peg,

$$\Delta e_t^r \approx \pi_t^f - \pi_t = 3.0 - 1.2 = 1.8\%.$$



Source: Statistics Denmark and Eurostat, April 2026.
Lower Danish inflation under a fixed nominal rate implies a real depreciation against the euro area, that is, an improvement in price competitiveness.

Long-Run Evidence: Relative Prices and Exchange Rates



Source: Sørensen & Whitta-Jacobsen, Figure 23.3.
Over long horizons, nominal exchange rates and relative prices tend to move together (RPPP). Short-run deviations can be large and persistent, especially in floating regimes.

Goods Market Equilibrium in an Open Economy

Add net exports to closed-economy demand:

$$\boxed{Y = C + I + G + NX} \qquad \text{where} \qquad NX = X - E^r M.$$

- X: export volume, measured in domestic goods.
- M: import volume, measured in foreign goods.
- $E^r M$: imports converted into domestic-goods units.
- The real exchange rate affects both export volumes and the cost of each imported unit.

Exports and Imports: The Sign Logic

Reduced-form summary, signs above each argument:

$$X = X\left(\overset{+}{E^r}, \overset{+}{Y^f}\right), \qquad M = M\left(\overset{-}{E^r}, \overset{+}{Y}, \overset{-}{\tau}, \overset{-}{r}, \overset{+}{\varepsilon}\right).$$

- Higher foreign income raises demand for domestic exports.
- Higher domestic income raises import demand.
- Higher taxes and higher real interest rates reduce domestic spending and imports.
- A real depreciation shifts demand toward domestic goods (volume effect).

The ambiguity sits in $NX = X - E^r M$: imports cost more in domestic goods even if their volume falls.

Marshall-Lerner Condition

A real depreciation, $E^r \uparrow$, has two opposing effects on net exports.

Force	Effect on NX	Intuition
Volume effect	positive	exports rise, import volumes fall
Price effect	negative	each imported unit costs more in domestic goods

Starting from balanced trade, depreciation improves NX if and only if

$$\boxed{\varepsilon_X + \varepsilon_M > 1}$$

where $\varepsilon_X, \varepsilon_M$ are the absolute export and import price elasticities. In the model this implies $\beta_1 > 0$.

Derivation

The J-Curve: Why the Timing Can Look Wrong

The Marshall-Lerner condition is usually a medium-run condition.

- Immediately after depreciation, quantities are pinned by contracts and habits.
- The price effect kicks in fast: imported goods become more expensive at once.
- Export and import volumes adjust only over months and quarters.
- The trade balance often **worsens first**, then improves: the J-curve.

Mechanism in one line: relative prices change immediately, quantities take time to respond.

Mid-Lecture Socrative Check

Three questions, all conceptual. Answer individually first, then we discuss.

Go to **gosocrative.com** and enter room code:

$\boxed{\text{MACROECONOMICSB}}$

Visit **gosocrative.com** and enter room
name MACROECONOMICSB

Three intuition questions follow. Pick the option that best captures the economic mechanism, not just the sign.

Socrative Q1: The Trilemma in Practice

Question. A small open economy keeps a credible fixed exchange rate against a large neighbour and allows free capital mobility. The neighbour's central bank announces a surprise rate cut. What is the most likely immediate response in the small economy?

(A) The domestic central bank raises rates to attract capital inflows.

(B) The domestic central bank cuts rates by the same amount to keep the peg credible.

(C) The exchange rate appreciates, since the foreign rate cut weakens the foreign currency.

(D) The domestic central bank holds rates unchanged but expands fiscal policy.

(E) Capital outflows force a one-time devaluation.

Socrative Q2: J-Curve Intuition

Question. A country that has lost competitiveness over many years finally devalues by 10%. In the first few months, the trade balance worsens slightly before it improves. What is the most plausible explanation?

(A) The Marshall-Lerner condition is permanently violated for this country.

(B) The country is too small for trade elasticities to matter.

(C) Export and import quantities adjust slowly while import prices rise immediately in domestic currency.

(D) The devaluation triggered higher domestic inflation that offset the gain.

(E) Imports stop entirely because consumers anticipate further devaluations.

Socrative Q3: Open-Economy Adjustment Under a Peg

Question. Denmark enters a mild recession while the euro area economy keeps growing as expected. The peg holds and no new policy is announced. Through which channel does Danish output return to long-run equilibrium in the open-economy model?

(A) The Danish central bank cuts its rate below the ECB to stimulate demand.

(B) The krone depreciates within the ERM II band until competitiveness is restored.

(C) Danish inflation falls below euro-area inflation, raising the real exchange rate and net exports.

(D) The Danish government must devalue the peg to restore competitiveness.

(E) Foreign demand automatically rises because the euro area absorbs the slack.

Open-Economy IS Curve

Linearising goods-market equilibrium around long-run values gives:

$$\boxed{y_t - \bar{y} = \beta_1(e_t^r - \bar{e}^r) - \beta_2(r_t - \bar{r}) + z_t}$$

- $\beta_1 > 0$: sensitivity of net exports to competitiveness; large if Marshall-Lerner holds strongly.
- $\beta_2 > 0$: sensitivity of private spending to the real interest rate.
- z_t: fiscal impulses, foreign-demand shocks, confidence shocks, sector-specific export booms.

We normalise $\bar{e}^r = 0$ in what follows.

Derivation sketch

From IS to the Preliminary AD Curve

Combine the IS curve with the real exchange rate dynamics

$$e^r_t = e^r_{t-1} + \Delta e_t + \pi^f_t - \pi_t.$$

Substituting into the IS curve gives the **preliminary AD curve**:

$$\boxed{y_t - \bar{y} = \beta_1\left(e^r_{t-1} + \Delta e_t + \pi^f_t - \pi_t\right) - \beta_2(r_t - \bar{r}) + z_t}$$

It is *preliminary* because Δe_t and r_t are not yet pinned down; that is the job of the exchange-rate regime, in lectures 10 and 11.

Full substitution

What the Two Demand Coefficients Mean

Coefficient	Larger when...	Effect
β_1	exports and imports react strongly to relative prices	real depreciation gives a larger AD boost
β_2	consumption and investment react strongly to the real interest rate	monetary tightening gives a larger AD contraction

The exchange-rate regime determines whether inflation moves demand mainly through e^r, through r, or through both:

- Fixed exchange rate: $\Delta e_t = 0$ and r_t is tied to r_t^f through UIP. Adjustment runs through inflation differentials and β_1.
- Floating exchange rate: monetary policy moves i_t and, through UIP, Δe_t. Both channels are active.

Aggregate Supply Is Mostly the Same

Opening the economy changes the demand side first. The AS curve is unchanged:

$$\boxed{\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t}$$

- Output above structural output raises wage and price pressure.
- A cost-push shock, $s_t > 0$, shifts AS upward.
- The position of AS depends on expected inflation.

For lectures 10 and 11, the exchange-rate regime determines AD; AS is the common building block.

Long-Run Equilibrium in the Open Economy

In long-run equilibrium:

$$\boxed{y_t = \bar{y}, \qquad \pi_t = \pi_t^e = \pi_t^f, \qquad \Delta e_t^r = 0}$$

- Output returns to structural output.
- Expected inflation matches actual inflation and matches foreign inflation.
- Relative purchasing power parity holds.
- Competitiveness is stable, so net exports are not drifting forever.

This adds a new long-run restriction compared with the closed economy: the real exchange rate cannot trend without bound.

Adjustment From a Recession: The Open-Economy Mechanism

Start with $y_t < \bar{y}$ and no new shock.

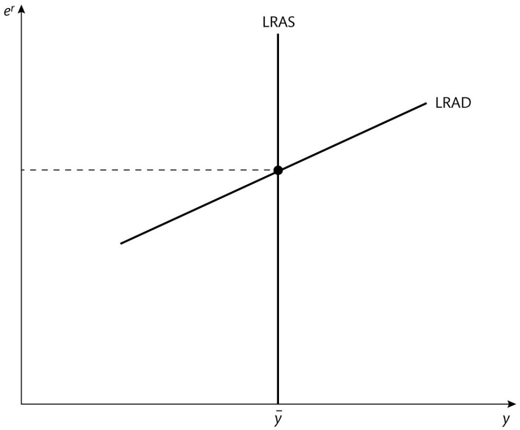
1. $y_t < \bar{y}$ lowers wage and price pressure: $\pi_t < \pi^f$.
2. With $\Delta e_t \approx 0$ under the peg, $\Delta e_t^r = \pi^f - \pi_t > 0$.
3. Competitiveness improves, so net exports rise.
4. AD shifts right and output moves back toward $\bar{y}$.

The open-economy propagation chain is:

$$\boxed{\text{recession} \;\rightarrow\; \text{low inflation} \;\rightarrow\; \text{real depreciation} \;\rightarrow\; \text{export-led recovery.}}$$

This is the answer to Socrative Q3.

Long-Run Aggregate Demand and Supply



Source: Sørensen & Whitta-Jacobsen, Figure 23.5.
The long-run equilibrium combines $y = \bar{y}$ with a real exchange rate consistent with stable competitiveness and balanced inflation differentials.

Denmark's Current Account Surplus: A Trade-Channel Country

Source: Sørensen & Whitta-Jacobsen, Figure 23.7; updated with Statistics Denmark.
Denmark has had a current-account surplus every year since 1990 except 1998. The surplus reached about 8.5% of GDP in 2024, making net exports central to Danish business-cycle analysis.

Punchline: Novo Nordisk and the Open-Economy Puzzle

Recent Danish growth has been dominated by a single firm.

- Novo Nordisk's sales were about 8.3% of Danish GDP in 2023, up from roughly 1% in the early 1990s.
- Pharmaceuticals reached about 24% of Danish goods exports in 2024.
- In Q3 2025, Danish GDP grew 2.3% quarter-on-quarter, the largest expansion since Q4 2021. Excluding pharma, growth would have been about 0.7%.

Through the model: a sustained sector-specific export boom is a persistent positive z_t shock; foreign demand y^f for one product category is exceptionally strong; the real exchange rate is pushed toward appreciation; the trilemma forces the Nationalbank to follow ECB rates rather than lean against the krone.

Source: IMF Article IV Country Report 2024 No. 293; Statistics Denmark, Q3 2025; CNBC, November 2025.

Same Shock, Different Regimes

Consider a negative demand shock that hits Denmark and Sweden symmetrically.

Country type	Monetary channel	Real-exchange-rate channel
Fixed exchange rate (Denmark)	policy rate constrained by the peg	adjustment mainly through inflation differentials
Floating exchange rate (Sweden)	policy rate can react to domestic conditions	nominal depreciation can move e^r quickly

The same open-economy equations apply to both. The exchange-rate regime determines which variables are free to move and how fast.

Notebook Exercise

Open `L9_open_economy_exchange_rates_trade.ipynb` and work through:

1. Compute the UIP-consistent domestic interest rate for a grid of expected depreciations.
2. Use Danish and euro-area HICP data to compute an approximate real-exchange-rate path under the peg.
3. Simulate a J-curve where Marshall-Lerner holds in the long run but quantities respond slowly.
4. Simulate the open-economy AD adjustment after a negative demand shock.

Sections 1 to 3 are for today. Sections 4 to 5 can be completed before the exercise class.

Takeaways

- UIP: $i \approx i^f + \Delta e^e$. Returns must be equal after expected currency movements.
- Trilemma: a country cannot have free capital mobility, a fixed exchange rate, and independent monetary policy at the same time.
- Real exchange rate: $\Delta e^r = \Delta e + \pi^f - \pi$. Competitiveness moves through nominal depreciation and inflation differentials.
- Marshall-Lerner: $\varepsilon_X + \varepsilon_M > 1$ is the condition under which real depreciation raises net exports.
- Open-economy IS: net exports add a second demand channel through β_1.
- Adjustment under a peg: recession $\rightarrow$ low inflation $\rightarrow$ real depreciation $\rightarrow$ export-led recovery.
- Danish reality: pharma exports and a 8.5% current-account surplus make the trade channel quantitatively central.

Preview of Lecture 10

Next time we specialise Chapter 23 to a fixed exchange rate (Denmark).

- We set $\Delta e_t = 0$ and $\Delta e^e_{t+1} = 0$.
- UIP then pins the domestic interest rate to the foreign rate.
- The real exchange rate adjusts only through inflation differentials.
- We compare unsystematic and systematic fiscal policy under the peg.
- Lecture 11 then specialises to a floating exchange rate (Sweden).

Appendix: UIP From Exact Returns to Logs

Start from no arbitrage:

$$1 + i_t = (1 + i_t^f)\frac{E^e_{t+1}}{E_t}.$$

Take logs of both sides:

$$\ln(1 + i_t) = \ln(1 + i_t^f) + \ln E^e_{t+1} - \ln E_t.$$

Apply $\ln(1 + x) \approx x$ for small x and use $e_t \equiv \ln E_t$:

$$i_t \approx i_t^f + e^e_{t+1} - e_t = i_t^f + \Delta e^e_{t+1}.$$

Reading. If the domestic currency is expected to depreciate, $\Delta e^e_{t+1} > 0$, investors require a higher domestic interest rate to be indifferent between bonds.

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Appendix: Real Exchange Rate Dynamics

Start from the level definition:

$$E_t^r = \frac{E_t P_t^f}{P_t}.$$

Take logs:

$$e_t^r = e_t + p_t^f - p_t.$$

Subtract the lagged equation:

$$\begin{aligned}\Delta e_t^r &= (e_t - e_{t-1}) + (p_t^f - p_{t-1}^f) - (p_t - p_{t-1}) \\ &= \Delta e_t + \pi_t^f - \pi_t.\end{aligned}$$

Long-run stability of competitiveness requires $\Delta e_t^r = 0$, so $\Delta e_t = \pi_t - \pi_t^f$. Under a fixed exchange rate, $\Delta e_t = 0$, so RPPP requires $\pi_t = \pi_t^f$.

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Appendix: Marshall-Lerner Derivation

Write net exports as

$$NX = X(E^r) - E^r M(E^r).$$

Differentiate with respect to E^r:

$$\frac{\partial NX}{\partial E^r} = X'(E^r) - M(E^r) - E^r M'(E^r).$$

Assume initially balanced trade: $X_0 = E_0^r M_0$. Define elasticities

$$\varepsilon_X \equiv \frac{E_0^r\, X'(E_0^r)}{X_0} > 0, \qquad \varepsilon_M \equiv -\frac{E_0^r\, M'(E_0^r)}{M_0} > 0.$$

Multiply through by E_0^r/X_0 and collect terms:

$$\frac{E_0^r}{X_0}\frac{\partial NX}{\partial E^r} = \varepsilon_X + \varepsilon_M - 1.$$

Hence $\partial NX/\partial E^r > 0$ if and only if $\varepsilon_X + \varepsilon_M > 1$.

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Appendix: Open-Economy IS Derivation Sketch

Start from goods-market equilibrium:

$$Y = C(Y, r) + I(Y, r) + G + X(E^r, Y^f) - E^r M(Y, E^r, r, \tau, \varepsilon).$$

Linearise around long-run values. Let $\alpha_D \in (0, 1)$ be the marginal propensity to spend on domestic goods. Collect terms in $y - \bar{y}$ on the left and solve:

$$y_t - \bar{y} = \beta_1(e_t^r - \bar{e}^r) - \beta_2(r_t - \bar{r}) + z_t,$$

where β_1 summarises the net-export response to competitiveness, weighted by trade shares and elasticities; β_2 summarises the absorption response to the real interest rate, with a Keynesian multiplier $1/(1 - \alpha_D)$; z_t collects fiscal, foreign-demand, and confidence shocks. Larger $\varepsilon_X + \varepsilon_M$ implies larger β_1.

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Appendix: Preliminary AD Substitution

Start from the open-economy IS curve with $\bar{e}^r = 0$:

$$y_t - \bar{y} = \beta_1 e_t^r - \beta_2(r_t - \bar{r}) + z_t.$$

Use the real-exchange-rate identity:

$$e_t^r = e_{t-1}^r + \Delta e_t + \pi_t^f - \pi_t.$$

Substitute step by step:

$$y_t - \bar{y} = \beta_1\left(e_{t-1}^r + \Delta e_t + \pi_t^f - \pi_t\right) - \beta_2(r_t - \bar{r}) + z_t.$$

- e_{t-1}^r: predetermined when period t starts.
- π_t: feeds back into demand through competitiveness (slope $-\beta_1$ in the (π, y) plane).
- r_t and Δe_t: pinned down by the exchange-rate regime, treated in lectures 10 and 11.

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